

Report — Task 1: Best Model Per Dataset Report

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cells.pt — Target: 90%

Model	Test Accuracy
AlexNet	94.91%
VGG16	96.14%
ResNet18	95.32%

Best model: VGG16, at 96.14% — crosses the 90% target.

Configuration Used:

```
{
  "DATA": "cells",
  "DATA_PATH": "data",
  "MODEL": "VGG16",
  "CHANNELS": 3,
  "NUM_CLASSES": 8,
  "BATCH_SIZE": 64,
  "EPOCHS": 15,
  "LEARNING_RATE": 0.001,
  "ACTIVATION": "ReLU",
  "DROP": 0.5
}
```

chest.pt — Target: 87%

Model	Test Accuracy
AlexNet	84.46%
VGG16	91.99%
ResNet18	81.25%

Best model: VGG16, at 91.99% — crosses the 87% target.

Configuration Used:

```
{
  "DATA": "chest",
  "DATA_PATH": "data",
  "MODEL": "VGG16",
  "CHANNELS": 1,
  "NUM_CLASSES": 2,
  "BATCH_SIZE": 64,
  "EPOCHS": 15,
  "LEARNING_RATE": 0.001,
  "ACTIVATION": "ReLU",
  "DROP": 0.5
}
```

lesions.pt — Target: 67%

Model	Test Accuracy
AlexNet	74.56%
VGG16	71.42%
ResNet18	73.97%

Best model: AlexNet, at 74.56% — crosses the 67% target.

Configuration Used:

```
{
  "DATA": "lesions",
  "DATA_PATH": "data",
  "MODEL": "AlexNet",
  "CHANNELS": 3,
  "NUM_CLASSES": 7,
  "BATCH_SIZE": 64,
  "EPOCHS": 15,
  "LEARNING_RATE": 0.001,
  "ACTIVATION": "ReLU",
  "DROP": 0.5
}
```

orgs.pt — Target: 83%

Model	Test Accuracy
AlexNet	89.14%
VGG16	90.21%
ResNet18	92.04%

Best model: ResNet18, at 92.04% — crosses the 83% target.

Configuration Used:

```
{
  "DATA": "orgs",
  "DATA_PATH": "data",
  "MODEL": "ResNet18",
  "CHANNELS": 1,
  "NUM_CLASSES": 11,
  "BATCH_SIZE": 64,
  "EPOCHS": 15,
  "LEARNING_RATE": 0.001,
  "ACTIVATION": "ReLU",
  "DROP": 0.5
}
```